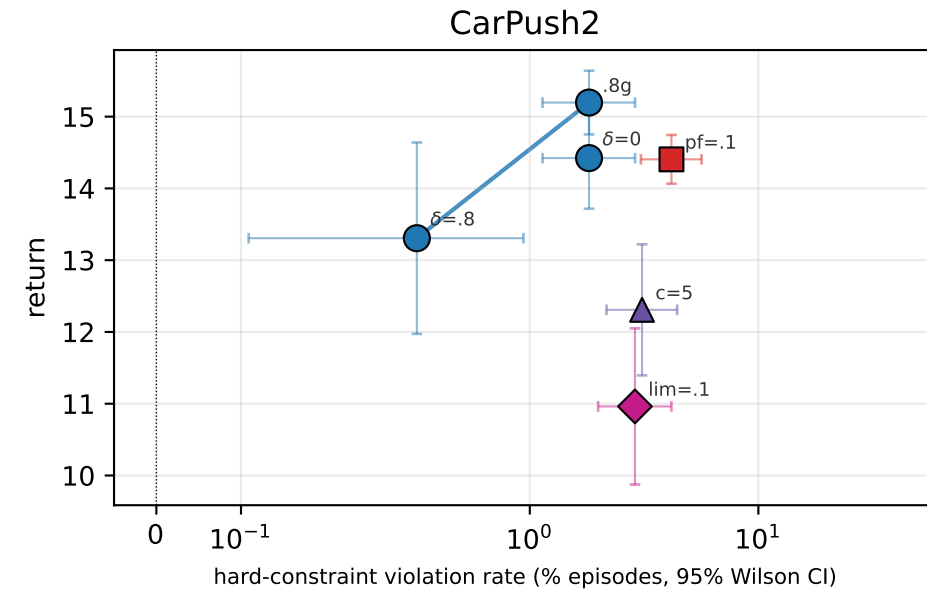
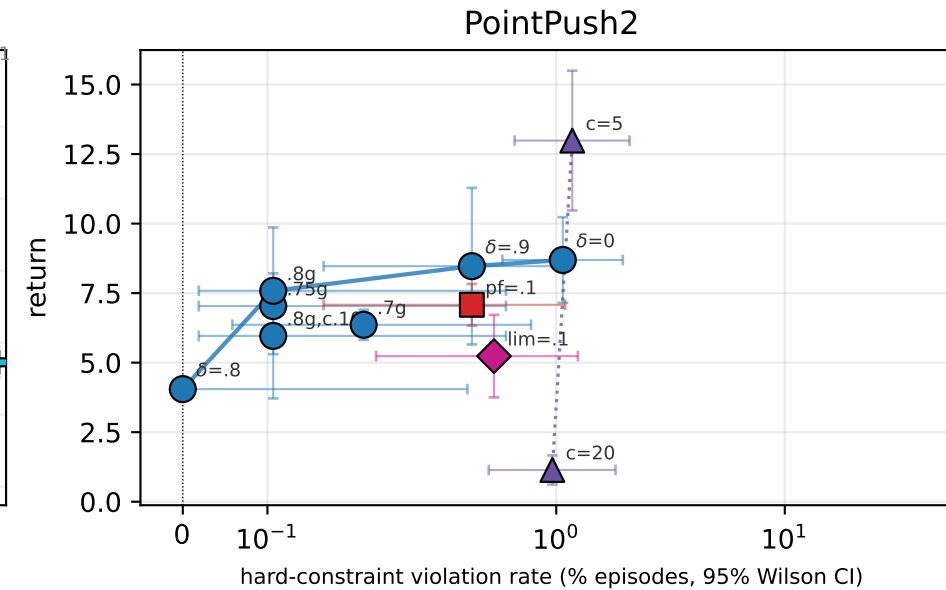
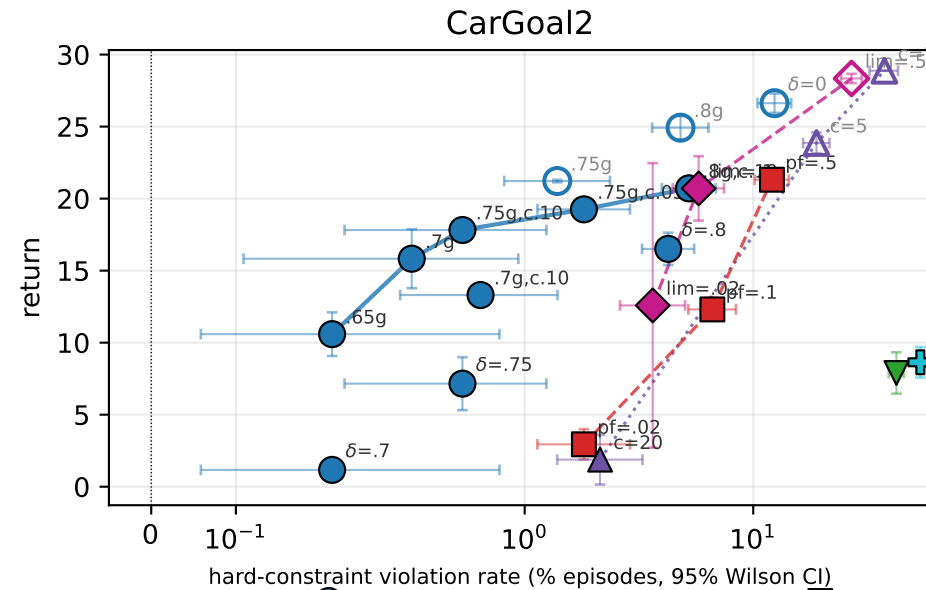
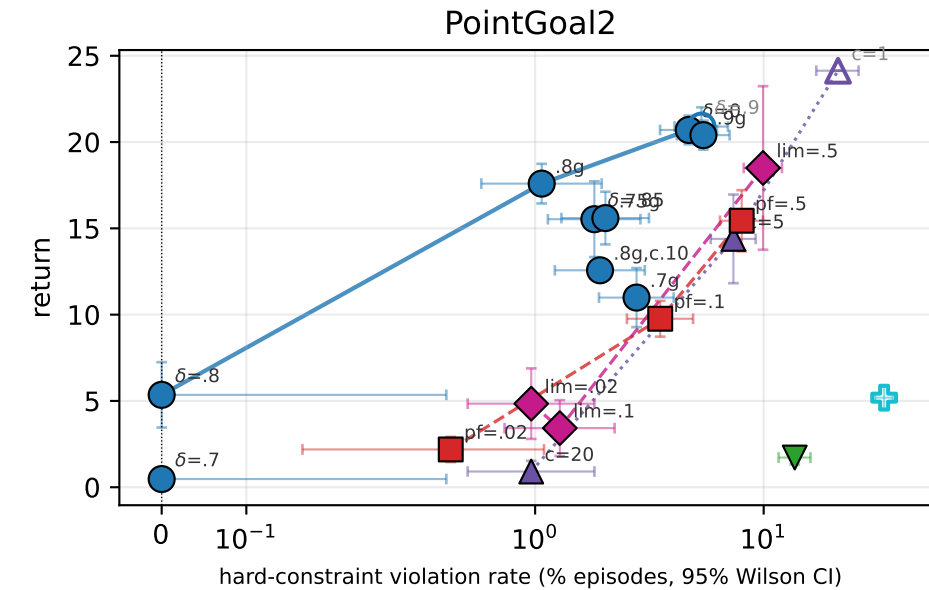


Return vs. violation rate (final checkpoint, 312 episodes/seed; hollow = soft cost > 25/episode)



 SafeOT-typed (ours)
  SDAC (multi-constraint SOTA)
  SAC-Pen coefficient sweep
 FDPI (constraint-based)
  SRCPO (multi-constraint SOTA)
  FDPI p_f sweep
 SAC-Lagrangian (penalty-based)
  typed Pareto frontier (soft ≤ 25)
  SAC-Lag cost-limit sweep
 SAC + fixed penalty (penalty-based)